

RONIT RAJ BAISHYA

Guwahati, Assam, India | ronitb@vt.edu | (540) 557-8814 | linkedin.com/in/ronitrajbaishya

RESEARCH PROFILE

Interests focus on the application of scientific machine learning and computational science for physical systems, mainly emphasis on neural operators, physics-informed neural networks for structural elements and dynamical systems, modeling, analysis, inverse problems, uncertainty quantification, and parameter/hidden state estimation. Broader interests in PDE-constrained optimization, design, estimation, and control. Experience in the application of machine learning, signal processing, statistical modeling, modal analysis, structural engineering, and dynamical systems.

EDUCATION

Lehigh University PhD in Structural Engineering	Bethlehem, Pennsylvania Aug 2025–Aug 2026
Virginia Polytechnic Institute and State University MS in Civil Engineering, Structural Engineering and Materials GPA: 3.62/4.00	Blacksburg, Virginia May 2024
Jorhat Engineering College Bachelor of Engineering in Civil Engineering GPA: 7.25/10	Jorhat, Assam, India Sept 2021

RELEVANT COURSES

ML/AI	Scientific Machine Learning, Data Mining, Statistics and Machine Learning, Machine Learning-I, Data Analytics, Time Series Analysis
System Dynamics	Nonlinear Systems and Chaos, Instrumentation and Signal Processing, Applied Linear Controls, Structural Dynamics
Civil and Structural Background	Structural Mechanics, Structural Analysis, Computer Methods of Structural Analysis, Mechanics of Composite Materials, Intermediate RC and Steel Design, Design of Prestressed Concrete, Fluid Mechanics, Hydrological Machines, Geotechnical Engineering

TECHNICAL SKILLS

Programming	Python, R, Jupyter notebooks, VS-Code, MATLAB, Mathematica, L ^A T _E X
ML	Supervised learning, unsupervised learning, PINN, neural operators, CNN, inverse problem, FNO, DeepONet, scikit-learn, pandas, NumPy, JAX, linear regression, classification, clustering, uncertainty quantification, benchmarking, hyperparameter tuning, model evaluation, inference
Computing	Time-series analysis, ARIMA, ACF/PACF, signal processing, FFT/DFT, feature engineering, statistical modeling, Kalman filtering, modal analysis
Deployment	Streamlit dashboards, Git/GitHub, reproducible scripts, technical documentation, visualization pipelines
Engineering	Abaqus, MathCAD, MIDAS, Revu Bluebeam, MicroStation, SAP 2000, ETABS, RISA-3D, STAAD.Pro, Response 2000, MS PowerPoint, MS Excel

PUBLICATIONS

Predicting Mechanical Properties of Glass-Epoxy Composites Using Supervised Learning <i>Journal of Reinforced Plastics and Composites</i> , Sage Journals DOI: 10.31224/07316844251318775	Feb 2025
India Flood Inventory: Creation of a Multi-Source National Geospatial Database to Facilitate Comprehensive Flood Research <i>Natural Hazards</i> , Springer	May 2021

PREPRINTS

Modal Identification and Environmental Effects on Natural Frequencies Using openLAB Research Bridge EngrXiv Archives (Submitted: <i>Journal of Advances in Structural Engineering</i> , Sage Journals) DOI: 10.31224/8196	Sept 2026
Gas Stove Activity Detection Using Raspberry Pi and Pi Camera: A Signal Processing Approach TechRxiv Archives DOI: 10.36227/techrxiv.172556906.62692261/v1	Sept 2024

TECHNICAL PROJECTS

Physics-Informed DeepONet for Uncertainty-Aware Composite Identification Sept 2026-Present

- Develop a validated 3D orthotropic glass-epoxy three-point-bending FEM model and use it to train a physics-informed DeepONet that reconstructs the structural response from material parameters.
- Then solve the sparse inverse problem to estimate key properties such as E_2, G_{12}, G_{13} , quantify their uncertainty, and determine which property most strongly governs the observed response.

Bathymetry-Conditioned CFO for 2D Shallow-Water Dynamics Aug 2026-Present

- So far, the CFO was extended to full-state 2D shallow-water dynamics $[h, hu, hv]$ using an FNO backbone, added an SWE physics loss, validated, and shown that PI-CFO improves physical consistency and mass conservation at a small cost in trajectory accuracy.
- Next, varying bathymetry $b(x, y)$ will be added, make the model terrain-conditioned and well-balanced, test unseen terrains and lake-at-rest preservation, and then study resolution transfer such as $32 \times 32 \rightarrow 64 \times 64$.

Environment-Compensated Structural Health Monitoring Streamlit App Mar 2026–May 2026

- Built a reproducible ML pipeline and Streamlit dashboard using openLAB bridge data to model temperature and solar-radiation effects in vibration-based monitoring.
- Implemented OLS, Ridge, and Lasso residual models to generate an Environment-Normalized Damage/Novelty Index for anomaly detection and false-alarm reduction.

Comparative Deep Learning Models for Image Classification May 2026

- Built and benchmarked MLP, CNN, and Vision Transformer models on MNIST and CIFAR-10 using PyTorch.
- Developed preprocessing, training, inference, ROC/F1 evaluation, and hyperparameter-tuning workflows for reproducible model comparison.

Clustering Analysis of Simulated and Gene-Expression Datasets Mar 2026

- Applied unsupervised learning to simulated and high-dimensional scientific datasets for pattern discovery and cluster interpretation.
- Evaluated clustering behavior under sparse-label conditions, building methods relevant to environmental monitoring, hydrologic systems, and sensor-driven anomaly detection.

Physics-Informed Neural Networks for openLAB Bridge Damage Detection Oct 2025–Dec 2025

- Developed a physics-informed ML framework using vibration data, modal-frequency features, and damage indices for structural damage detection.
- Benchmarked the PINN approach against Random Forest, Ridge Regression, and Gradient Boosting to evaluate accuracy, interpretability, and generalization.
- Connected physical constraints with data-driven inference to support reliable ML models for civil and mechanical systems with limited labeled data.

Forecasting and Trend Analysis of Wind Speed and Wind-Power Generation Apr 2024–May 2024

- Modeled wind-speed and power-generation trends using ACF, PACF, cross-correlation, and ARIMA-based time-series forecasting.
- Developed forecasting skills transferable to environmental systems, water-resource modeling, renewable-energy planning, and climate-sensitive infrastructure.

Periodicity and Nonlinear Behavior of Lorenz Attractors Apr 2024–May 2024

- Analyzed Lorenz-system behavior under varying ρ values to study nonlinear dynamics, sensitivity, and chaos in physical systems.

Development of an Output Feedback Controller Apr 2024–May 2024

- Designed a discrete-time LQR output-feedback controller with Kalman filtering for state estimation under incomplete measurements.

RESEARCH, AND TEACHING EXPERIENCE

Indian Institute of Technology, Guwahati

Research Intern

Guwahati, Assam
August 2026–July 2027 (expected)

Lehigh University

Graduate Research Assistant

Bethlehem, Pennsylvania
May 2026–Aug 2026

Lehigh University

Graduate Teaching Assistant

Bethlehem, Pennsylvania
Jan 2026–May 2026

Virginia Tech

Blacksburg, Virginia

<i>Teaching Assistant</i>	Dec 2023–May 2024
Virginia Tech Vibration Testing Laboratory	Blacksburg, Virginia
<i>Graduate Assistant</i>	Sept 2022–Dec 2022
HydroSense Lab, Indian Institute of Technology, Delhi	Delhi, India
<i>Research Intern</i>	July 2020–May 2021

INDUSTRY EXPERIENCE

Kiewit Corporation	Lone Tree, Colorado
<i>Structural Engineer</i>	June 2024–Apr 2025
Kiewit Corporation	Lone Tree, Colorado
<i>Structural Engineer Intern</i>	May 2023–Aug 2023
Architects Collaboratives	Assam, India
<i>Graduate Engineer Trainee</i>	Aug 2021–July 2022

AWARDS AND FELLOWSHIPS

Kanai and Bibha Mukherjee Opportunity Fund , Lehigh University	May 2025
Rossin College Fellowship , Lehigh University	June 2025
Gibson Fellowship , Lehigh University	Nov 2025